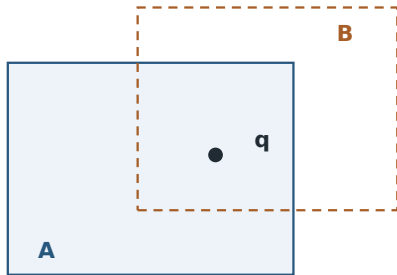


Result obligations and traversal behavior

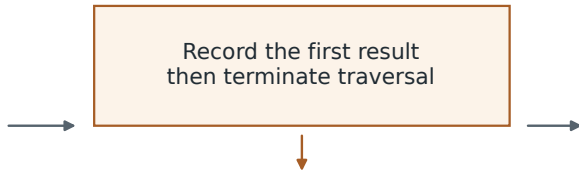
Illustration: a complete relation must retain both qualifying pairs

(a) Application meaning



q lies in A and B
Required: $\{(q,A), (q,B)\}$

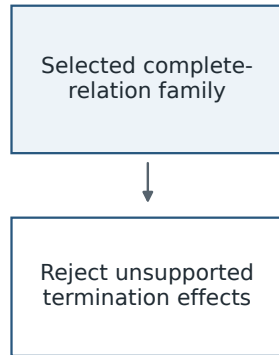
(b) A legal RT action can conflict



The second qualifying pair
may never be returned.

First-hit order is unspecified.

(c) RTDL boundary

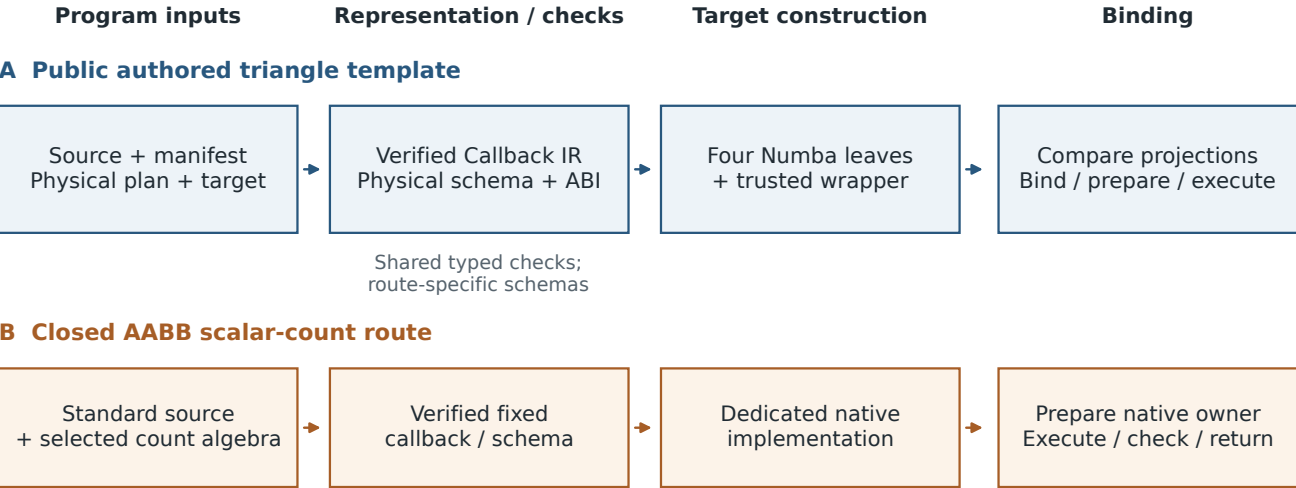


This illustrates a fixed result-family rule; it does not infer the application-to-RT mapping.

RTDL language and implementation routes

Authored code generation and closed native routes have distinct compilation boundaries

Author retains: application algorithm, RT mapping, metadata meaning, intended output



Runtime: prepared owners + route-specific identity, status and return checks.

Trusted components: topology wrappers, specialized native targets and runtime providers.

An authored role and its physical execution

Nearest-hit triangle template: logical closest-hit / miss are called from the raygen wrapper

(a) Author-supplied computation

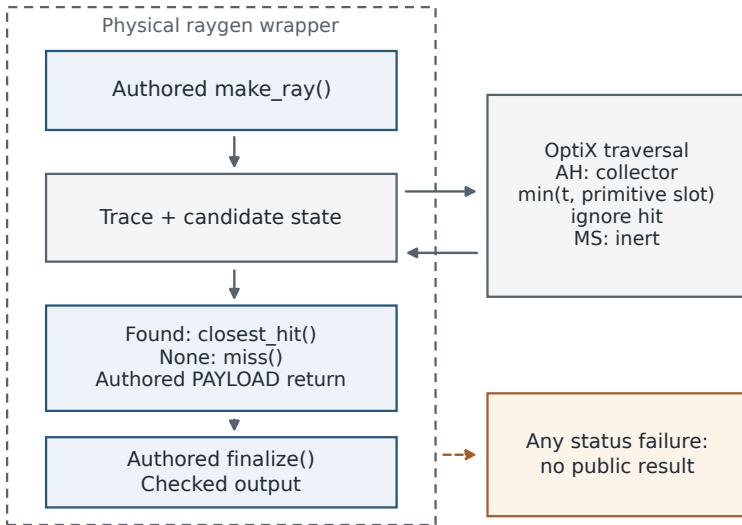
```
primitive_id = first_metadata[
    hit.primitive_index]
```

Closest-hit expression excerpt

Mesh slots	0	1
First metadata	17	29
Second metadata	31	43

```
Hit slot 0 → (17, 1, HIT_TAG)
Hit slot 1 → (29, 1, HIT_TAG)
Miss → (U32_MAX, 0, MISS_TAG)
```

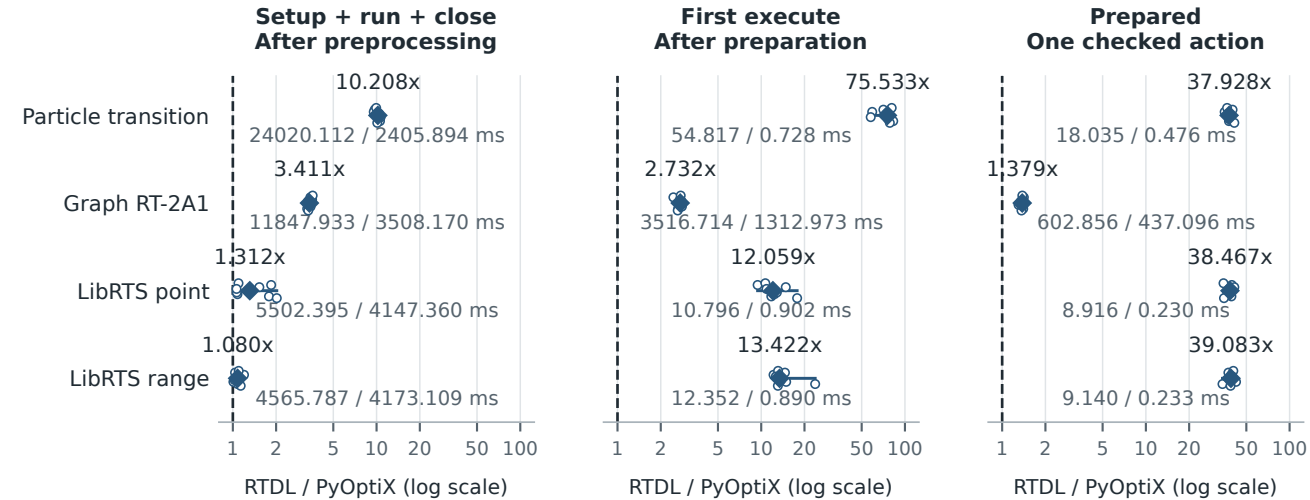
(b) Compiler-owned execution



Source-traced template; expected outputs shown. This figure adds no GPU execution evidence.

Retained application costs relative to public PyOptiX

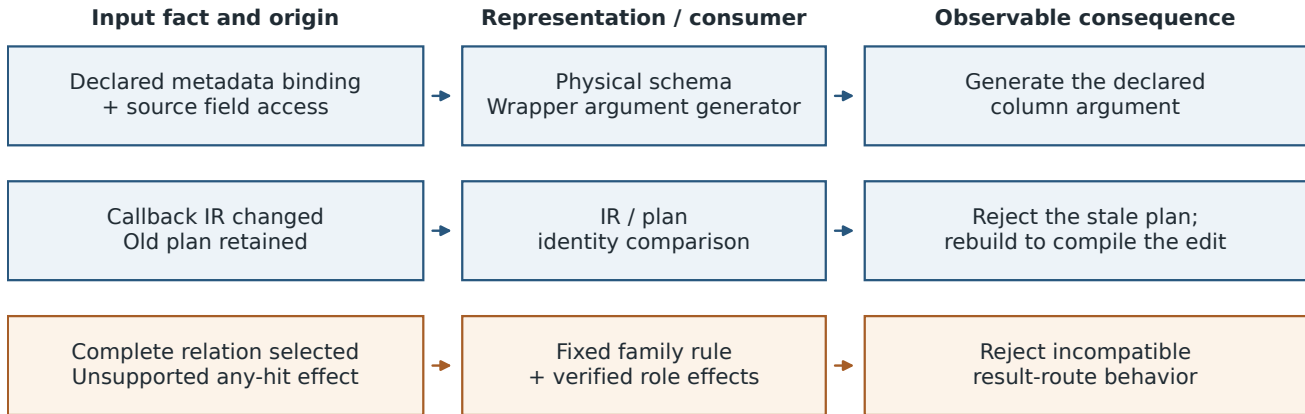
RTX A4500 | 3 applications, 4 operation units | before the current remediation



Diamond: median paired ratio. Line: observed 8-block range, not a confidence interval.
Open circles: all blocks. Small labels: separate RTDL / PyOptiX time medians in ms.
1x means equal time; every retained block is above 1x. New long-workload results are not included.

Program facts consumed by the compiler

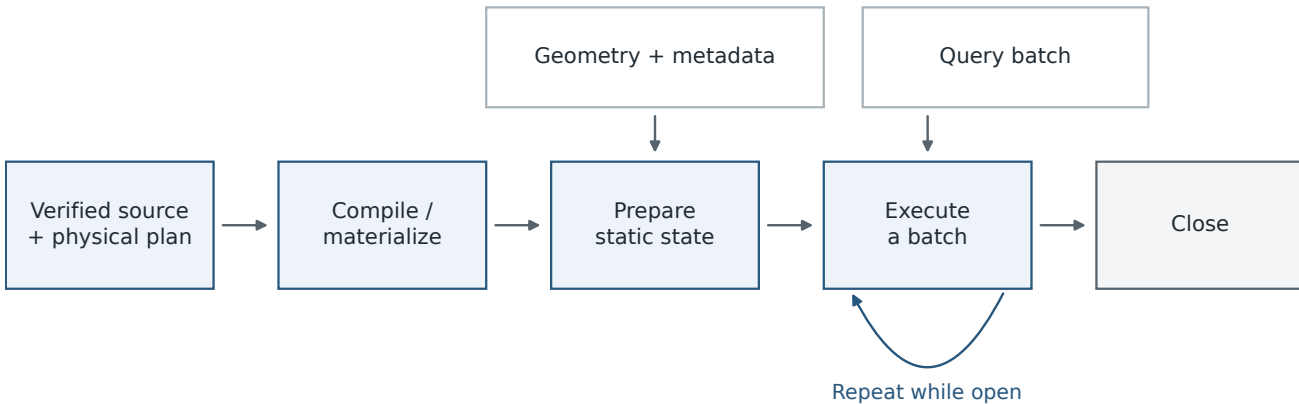
A representation matters where its facts change checking, generation or selection



First two rows: authored triangle. Third row: a distinct fixed complete-relation route.

Prepared-program lifecycle and input ownership

Public authored triangle interface; static inputs and batch inputs enter at different boundaries



Static/batch separation belongs to the physical plan and lifecycle, not a manifest field.

Application mappings and selected implementation routes

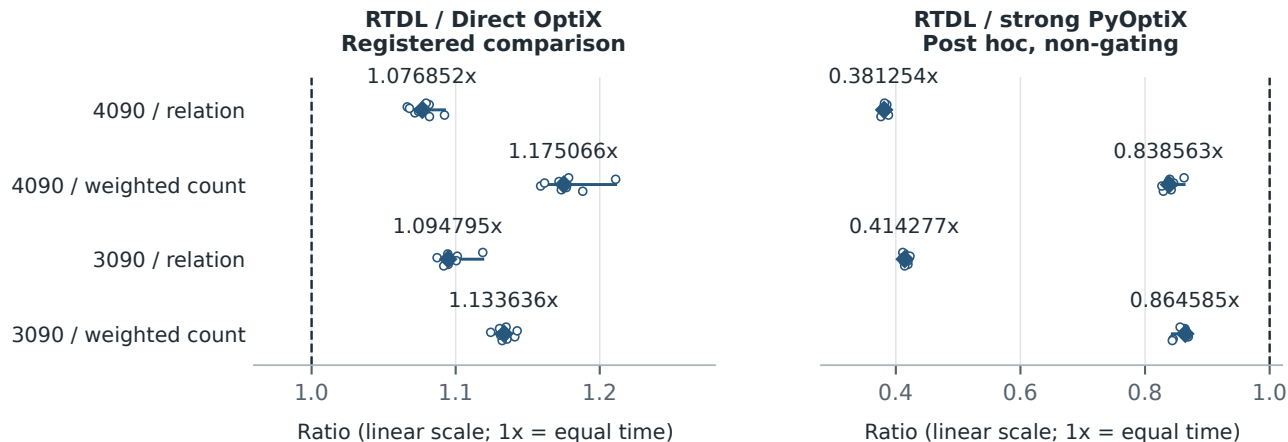
Source-traced route classes and the retained three-application PyOptiX study

Application	Selected route	A/C measured stage
Particle	Generated role leaves	One transition
Graph triangle counting	Generated role leaves	Whole-graph RT-2A1
LibRTS	Fixed native count	Point + range counts
RayDB	Fixed native partner	Not measured
X-HD	Fixed native partner	Not measured
RTNN	Fixed native partner	Not measured
RT-DBSCAN	Fixed native / continuation	Not measured
Spatial RayJoin	Fixed native partner	Not measured
RT-BarnesHut	Hierarchy plan + native partner	Not measured

Nine mappings; three measured applications / four operation units. No claim of nine current benchmarks.

Prepared costs of two restricted standard tasks

Independent RTX 4090 / RTX 3090 transactions; separate from the application campaign



Eight paired blocks per row; lines are observed ranges, not confidence intervals.

Only two tuned tasks. All four post-import rows remain adverse (worst block: 2.377129x).

Both GPUs: 4,096 timed A calls; 32 separate detailed diagnostic receipts. Paired overhead: A only.

These prepared ratios do not establish nine-application performance or lifecycle-independent language speed.